

Continuum Mechanics

class # 06

09/03/2026

Tensor Calculus

Definition (2.1) By the standard Euclidean norms on $\mathbb{E}^3$, $\mathcal{V}$, and $\mathcal{V}^2$ we mean

$$|\vec{x}| = \sqrt{\vec{x} \cdot \vec{x}}, \quad \forall \vec{x} \in \mathbb{E}^3$$

$$|\vec{v}| = \sqrt{\vec{v} \cdot \vec{v}}, \quad \forall \vec{v} \in \mathcal{V}$$

$$|S| = \sqrt{S:S}, \quad \forall S \in \mathcal{V}^2$$

Definition (2.2) Consider a function $f: \mathcal{V} \rightarrow W$, where W denotes any of the spaces $\mathbb{E}^3$, $\mathcal{V}$, $\mathcal{V}^2$ or $\mathbb{R}$.

(i) If there are constants $C > 0$ and $r > 0$ such that

$$|f(\vec{u})| \leq C |\vec{u}|^r$$

as $\vec{u} \rightarrow \vec{0}$, then we write

$$f(\vec{u}) = O(|\vec{u}|^r), \quad \text{as } \vec{u} \rightarrow \vec{0}$$

(ii) If there is a constant $r > 0$ such that

$$\frac{|f(\vec{u})|}{|\vec{u}|^r} \rightarrow 0 \quad \text{as } \vec{u} \rightarrow \vec{0}$$

then we write

$$f(\vec{u}) = o(|\vec{u}|^r), \quad \text{as } \vec{u} \rightarrow \vec{0}.$$

The symbols O and o are called order symbols.

Proposition: If $f(\vec{u}) = o(|\vec{u}|^r)$ as $\vec{u} \rightarrow 0$ implies $f(\vec{u}) = O(|\vec{u}|^r)$ as $\vec{u} \rightarrow 0$.

Proof: Suppose that $f(\vec{u}) = o(|\vec{u}|^r)$, as $\vec{u} \rightarrow 0$.
We need to know what this means specifically.

For every $\alpha > 0$, there is a $\delta > 0$, such that if

$$|\vec{u}| < \delta$$

then

$$\frac{|f(\vec{u})|}{|\vec{u}|^r} < \alpha$$

So, fix $\alpha > 0$. Then, if $|\vec{u}| < \delta$, we have

$$|f(\vec{u})| < \alpha |\vec{u}|^r.$$

Thus $f(\vec{u}) = O(|\vec{u}|^r)$, taking $C = \alpha$. //

Proposition: If $f(\vec{u}) = O(|\vec{u}|^p)$ as $\vec{u} \rightarrow 0$,
and $p > r > 0$. Then $f(\vec{u}) = o(|\vec{u}|^r)$.

Proof: Exercise.

Remark: The statement $f(\vec{u}) = o(|\vec{u}|^r)$, as $\vec{u} \rightarrow 0$,
means $|f(\vec{u})|$ goes to zero faster than $|\vec{u}|^r$.

Definition: Given two functions $f, g: \mathcal{V} \rightarrow W$, the notation

$$f(\vec{u}) = g(\vec{u}) + \mathcal{O}(|\vec{u}|^r), \text{ as } \vec{u} \rightarrow 0$$

means

$$f(\vec{u}) - g(\vec{u}) = \mathcal{O}(|\vec{u}|^r), \text{ as } \vec{u} \rightarrow 0.$$

Similarly,

$$f(\vec{u}) = g(\vec{u}) + o(|\vec{u}|^r), \text{ as } \vec{u} \rightarrow 0$$

means

$$f(\vec{u}) - g(\vec{u}) = o(|\vec{u}|^r), \text{ as } \vec{u} \rightarrow 0.$$

Definition: Let $\Omega \subseteq \mathbb{E}^3$. A function $\phi: \Omega \rightarrow \mathbb{R}$ is called a scalar field. A function $\vec{g}: \Omega \rightarrow \mathcal{V}$ is called a vector field. A function $S: \Omega \rightarrow \mathcal{V}^2$ is called a tensor field.

Definition (2.3): A scalar field $\phi: \mathbb{E}^3 \rightarrow \mathbb{R}$ is said to be differentiable at $\vec{x} \in \mathbb{E}^3$ iff there exists a vector $\nabla \phi(\vec{x}) \in \mathcal{V}$ such that

$$\phi(\vec{x} + \vec{h}) = \phi(\vec{x}) + \nabla \phi(\vec{x}) \cdot \vec{h} + o(|h|),$$

as $\vec{h} \rightarrow 0$. Equivalently

$$\nabla \phi(\vec{x}) \cdot \vec{a} = \left. \frac{d}{ds} \phi(\vec{x} + s\vec{a}) \right|_{s=0}, \quad \forall \vec{a} \in \mathcal{V},$$

where $s \in \mathbb{R}$. The vector $\nabla \phi(\vec{x})$ is called the gradient of ϕ at $\vec{x}$.

Proposition: let $\phi: \mathbb{E}^3 \rightarrow \mathbb{R}$ be differentiable at $\vec{x} \in \mathbb{E}$. Then $\nabla \phi(\vec{x})$ is unique and the ascribed equivalence in the last definition holds.

Proof: See any good book on advanced calculus. //

Results (2.4): let $\mathcal{F} = \{\vec{e}_i\}$ be a frame. Then

$$\nabla \phi(\vec{x}) = \frac{\partial \phi}{\partial x_i}(\vec{x}) \vec{e}_i$$

where $\vec{x} \in \mathbb{E}^3$ is represented in the coordinate frame $\mathcal{F}$ as

$$\vec{x} = x_i \vec{e}_i.$$

Proof: Suppose that $\vec{a} \in V$ and

$$\vec{a} = a_i \vec{e}_i$$

For any $s \in \mathbb{R}$,

$$\phi(\vec{x} + s\vec{a}) = \phi(x_1 + sa_1, x_2 + sa_2, x_3 + sa_3)$$

Using the chain rule

$$\begin{aligned} \nabla \phi(\vec{x}) \cdot \vec{a} &= \frac{d}{ds} \phi(\vec{x} + s\vec{a})_{s=0} \\ &= \left[\frac{\partial \phi}{\partial x_1}(\vec{x} + s\vec{a}) a_1 + \frac{\partial \phi}{\partial x_2}(\vec{x} + s\vec{a}) a_2 \right. \\ &\quad \left. + \frac{\partial \phi}{\partial x_3}(\vec{x} + s\vec{a}) a_3 \right]_{s=0} \\ &= \frac{\partial \phi}{\partial x_i}(\vec{x}) a_i \end{aligned}$$

$$= \left(\frac{\partial \phi}{\partial x_j}(\vec{x}) \vec{e}_j \right) \cdot \vec{a}$$

Since $\vec{a} \in V$ is arbitrary,

$$\nabla \phi(\vec{x}) = \frac{\partial \phi}{\partial x_j}(\vec{x}) \vec{e}_j \quad |||$$

Remark: (1) The derivative of a scalar field $\phi: \mathbb{E}^3 \rightarrow \mathbb{R}$ is a vector field $\nabla \phi: \mathbb{E}^3 \rightarrow V$

(2) We will often use the notation

$$\phi_{,i} := \frac{\partial \phi}{\partial x_i}$$

Proposition: Suppose that $\phi: \mathbb{E}^3 \rightarrow \mathbb{R}$ is of class $C^1(\Omega; \mathbb{R})$, Ω is a bounded, open, convex subset of $\mathbb{E}^3$. Assume also that

$$|\nabla \phi(\vec{x}) - \nabla \phi(\vec{y})| \leq \gamma |\vec{x} - \vec{y}|$$

for some $\gamma > 0$ and all $\vec{x}, \vec{y} \in \Omega$. Then for all $\vec{x}, \vec{y} \in \Omega$,

$$|\phi(\vec{x}) - \phi(\vec{y}) - \nabla \phi(\vec{y}) \cdot (\vec{x} - \vec{y})| \leq \frac{\gamma}{2} |\vec{x} - \vec{y}|^2$$

Proof: A good book on Advanced Calculus. |||

let's write this in a more familiar way. Set

$$\vec{h} = \vec{x} - \vec{y} \in V$$

Then

$$\vec{x} = \vec{y} + \vec{h}$$

So

$$|\phi(\vec{y} + \vec{h}) - \phi(\vec{y}) - \nabla \phi(\vec{y}) \cdot \vec{h}| \leq \frac{\gamma}{2} |\vec{h}|^2$$

Hence

$$\phi(\vec{y} + \vec{h}) = \phi(\vec{y}) + \nabla \phi(\vec{y}) \cdot \vec{h} + O(|\vec{h}|^2).$$

Or, if you like to use $\vec{x}$ instead of $\vec{y}$,

$$\phi(\vec{x} + \vec{h}) = \phi(\vec{x}) + \nabla \phi(\vec{x}) \cdot \vec{h} + O(|\vec{h}|^2).$$

Definition (2.5): A vector field $\vec{v}: \mathbb{E}^3 \rightarrow \mathcal{V}$ is said to be differentiable at $\vec{x} \in \mathbb{E}^3$ iff there exists a second-order tensor $\nabla \vec{v} \in \mathcal{V}^2$ such that

$$\vec{v}(\vec{x} + \vec{h}) = \vec{v}(\vec{x}) + \nabla \vec{v}(\vec{x}) \vec{h} + o(|\vec{h}|).$$

or, equivalently,

$$\nabla \vec{v}(\vec{x}) \vec{a} = \left. \frac{d}{d\alpha} \vec{v}(\vec{x} + \alpha \vec{a}) \right|_{\alpha=0}, \quad \forall \vec{a} \in \mathcal{V},$$

where $\alpha \in \mathbb{R}$. The tensor $\nabla \vec{v}(\vec{x})$ is called the gradient of $\vec{v}$ at $\vec{x}$.

Result (2.6): Let $\mathcal{F} = \{\vec{e}_i\}$ be a frame and suppose

$$\vec{v}(\vec{x}) = v_i(\vec{x}) \vec{e}_i$$

Then

$$\nabla \vec{v}(\vec{x}) = \frac{\partial v_i}{\partial x_j}(\vec{x}) \vec{e}_i \otimes \vec{e}_j$$

where

$$\vec{x} = x_i \vec{e}_i.$$

Proof: For any vector $a = a_k e_k \in V$

$$v_i(\vec{x} + \alpha \vec{a}) = v_i(x_1 + \alpha a_1, x_2 + \alpha a_2, x_3 + \alpha a_3)$$

By the chain rule

$$\begin{aligned} \frac{d}{d\alpha} v_i(\vec{x} + \alpha \vec{a}) \Big|_{\alpha=0} &= \left[\frac{\partial v_i}{\partial x_1}(\vec{x} + \alpha \vec{a}) a_1 + \frac{\partial v_i}{\partial x_2}(\vec{x} + \alpha \vec{a}) a_2 \right. \\ &\quad \left. + \frac{\partial v_i}{\partial x_3}(\vec{x} + \alpha \vec{a}) a_3 \right]_{\alpha=0} \\ &= \frac{\partial v_i}{\partial x_j}(\vec{x}) a_j. \end{aligned}$$

Thus,

$$\begin{aligned} \nabla \vec{v}(\vec{x}) \vec{a} &= \frac{d}{d\alpha} \vec{v}(\vec{x} + \alpha \vec{a}) \Big|_{\alpha=0} \\ &= \frac{d}{d\alpha} v_i(\vec{x} + \alpha \vec{a}) \Big|_{\alpha=0} \vec{e}_i \\ &= \frac{\partial v_i}{\partial x_j}(\vec{x}) a_j \vec{e}_i \\ &= \left(\frac{\partial v_i}{\partial x_j}(\vec{x}) \vec{e}_i \otimes \vec{e}_j \right) \vec{a} \end{aligned}$$

Thus

$$\nabla \vec{v}(\vec{x}) = \frac{\partial v_i}{\partial x_j}(\vec{x}) \vec{e}_i \otimes \vec{e}_j. \quad ///$$

Remark: (1) The derivative of a vector field $\vec{v}: \mathbb{E}^3 \rightarrow V$ is a second-order tensor field $\nabla \vec{v}: \mathbb{E}^3 \rightarrow V^2$.

(2) As for scalar fields, we will often write

$$v_{i,j} := \frac{\partial v_i}{\partial x_j}$$

Proposition: Assume that $\Omega \subset \mathbb{E}^3$ is an open, bounded, convex set. Suppose that $\vec{v}: \mathbb{E}^3 \rightarrow \mathcal{V}$ satisfies

$$\vec{v} \in C^1(\Omega; \mathcal{V})$$

Assume that there is some $\gamma > 0$ such that

$$|\nabla \vec{v}(\vec{x}) - \nabla \vec{v}(\vec{y})| \leq \gamma |\vec{x} - \vec{y}|,$$

for all $\vec{x}, \vec{y} \in \Omega$. Then

$$|\vec{v}(\vec{x}) - \vec{v}(\vec{y}) - \nabla \vec{v}(\vec{y})(\vec{x} - \vec{y})| < \frac{\gamma}{2} |\vec{x} - \vec{y}|^2,$$

for all $\vec{x}, \vec{y} \in \Omega$.

Proof: See a good book on Advanced Calculus. ///

As before, we can use the last result to write

$$\vec{v}(\vec{x} + \vec{h}) = \vec{v}(\vec{x}) + \nabla \vec{v}(\vec{x})\vec{h} + \mathcal{O}(|\vec{h}|^2)$$

as $\vec{h} \rightarrow \vec{0}$.

Divergence

Definition (2.7): To any vector field $\vec{v}: \mathbb{E}^3 \rightarrow \mathcal{V}$ we associate a scalar field $\nabla \cdot \vec{v}: \mathbb{E}^3 \rightarrow \mathbb{R}$ defined by

$$\nabla \cdot \vec{v} = \text{tr}(\nabla \vec{v}).$$

$\nabla \cdot \vec{v}$ is called the divergence of $\vec{v}$.

Result (2.8): Let $\mathcal{F} = \{\vec{e}_i\}$ be a frame. Write

$$\vec{v}(\vec{x}) = v_i(\vec{x}) \vec{e}_i.$$

and assume that $\vec{v}: \mathbb{E}^3 \rightarrow \mathcal{V}$ is differentiable. Then,

$$\nabla \cdot \vec{v}(\vec{x}) = \frac{\partial v_i}{\partial x_i}(\vec{x}) = v_{i,i}(\vec{x}).$$

Proof: Exercise. ///

Definition (2.9): To any second-order tensor field $S: \mathbb{E}^3 \rightarrow \mathcal{V}^2$ that is differentiable we associate a vector field $\nabla \cdot S: \mathbb{E}^3 \rightarrow \mathcal{V}$ defined by the relation

$$(\nabla \cdot S) \cdot \vec{a} = \nabla \cdot (S^T \vec{a}),$$

for all constant vectors $\vec{a} \in \mathcal{V}$. We call $\nabla \cdot S$ the divergence of S .

Result (2.10): Let $\mathcal{F} = \{\vec{e}_i\}$ be a frame and assume that $S: \mathbb{E}^3 \rightarrow \mathcal{V}^2$ is differentiable. If

$$S = S_{ij} \vec{e}_i \otimes \vec{e}_j$$

then

$$(\nabla \cdot S)(\vec{x}) = \frac{\partial S_{ij}}{\partial x_j}(\vec{x}) \vec{e}_i = S_{ij,j}(\vec{x}) \vec{e}_i,$$

where

$$\vec{x} = x_i \vec{e}_i$$

Proof: let $\vec{a} = a_k \vec{e}_k \in \mathcal{V}$ and set

$$\vec{q} := S^T \vec{a}.$$

Write $\vec{q} = q_j \vec{e}_j$. Then

$$q_j = S_{ij} a_i.$$

Using Definition (2.9) and Result (2.8), we have

$$\begin{aligned} (\nabla \cdot S) \cdot \vec{a} &= \nabla \cdot (S^T \vec{a}) \\ &= \nabla \cdot \vec{q} \\ &= q_{j,j} \\ &= \frac{\partial q_j}{\partial x_j} \\ &= \frac{\partial}{\partial x_j} (S_{ij} a_i) \\ &= \frac{\partial S_{ij}}{\partial x_j} a_i \\ &= S_{ij,j} a_i \\ &= (S_{ij,j} \vec{e}_i) \cdot \vec{a} \end{aligned}$$

Therefore

$$\nabla \cdot S = S_{ij,j} \vec{e}_i \quad |||$$

Result (2.11): Let ϕ , $\vec{v}$, S be differentiable scalar, vector, and tensor fields, respectively. Then

$$\nabla(\phi \vec{v}) = \vec{v} \otimes \nabla \phi + \phi \nabla \vec{v}$$

$$\nabla \cdot (\phi S) = \phi \nabla \cdot S + S \nabla \phi$$

$$\nabla \cdot (S^T \vec{v}) = (\nabla \cdot S) \vec{v} + S : \nabla \vec{v}$$

Proof: Let $\mathcal{F} = \{\vec{e}_i\}$ be a frame. Set

$$\vec{v} = v_i \vec{e}_i, \quad S = S_{ij} \vec{e}_i \otimes \vec{e}_j.$$

Then, using Result 2.6,

$$\begin{aligned} \nabla(\phi \vec{v}) &= \frac{\partial(\phi v_i)}{\partial x_j} \vec{e}_i \otimes \vec{e}_j \\ &= \frac{\partial \phi}{\partial x_j} v_i \vec{e}_i \otimes \vec{e}_j + \phi \frac{\partial v_i}{\partial x_j} \vec{e}_i \otimes \vec{e}_j \\ &= (\vec{v}_i \cdot \vec{e}_i) \otimes \left(\frac{\partial \phi}{\partial x_j} \vec{e}_j \right) + \phi \nabla \vec{v} \\ &= \vec{v} \otimes \nabla \phi + \phi \nabla \vec{v}. \quad // \end{aligned}$$

Next, ϕS is a second-order tensor. Thus, using Result 2.10,

$$\begin{aligned} \nabla \cdot (\phi S) &= \frac{\partial(\phi S_{ij})}{\partial x_j} \vec{e}_i \\ &= \left(\frac{\partial \phi}{\partial x_j} S_{ij} + \phi \frac{\partial S_{ij}}{\partial x_j} \right) \vec{e}_i \\ &= S_{ij} \frac{\partial \phi}{\partial x_j} \vec{e}_i + \phi \nabla \cdot S \\ &= S \nabla \phi + \phi \nabla \cdot S. \quad // \end{aligned}$$

Finally, observe that $S^T \vec{v}$ is a vector field, and, in coordinate form,

$$\begin{aligned} S^T \vec{v} &= [S^T]_{ij} v_j \vec{e}_i \\ &= S_{ji} v_j \vec{e}_i \\ &= S_{ij} v_i \vec{e}_j \end{aligned}$$

Using Result 2.8, we have

$$\begin{aligned}\nabla \cdot (S^T \vec{v}) &= \frac{\partial (S_{ij} v_i)}{\partial x_j} \\ &= \frac{\partial S_{ij}}{\partial x_j} v_i + S_{ij} \frac{\partial v_i}{\partial x_j} \\ &= (\nabla \cdot S) \cdot \vec{v} + S : \nabla v\end{aligned}$$

Example (Exercise 2.4) let ϕ be a scalar field, $\vec{v}, \vec{w} \in \mathcal{V}$ be vector fields, and S be a second-order tensor field. Then

$$(a) \quad \nabla \cdot (\phi \vec{v}) = \nabla \phi \cdot \vec{v} + \phi \nabla \cdot \vec{v}$$

$$(b) \quad \nabla (\vec{v} \cdot \vec{w}) = (\nabla v)^T \vec{w} + (\nabla w)^T \vec{v}$$

$$(c) \quad \nabla \cdot (\vec{v} \otimes \vec{w}) = (\nabla v) \vec{w} + (\nabla \cdot \vec{w}) \vec{v}$$

Solution: (a):

$$\begin{aligned}\nabla \cdot (\phi \vec{v}) &= \frac{\partial}{\partial x_i} (\phi v_i) \\ &= \frac{\partial \phi}{\partial x_i} v_i + \phi \frac{\partial v_i}{\partial x_i} \\ &= \nabla \phi \cdot \vec{v} + \phi (\nabla \cdot \vec{v}) \quad //\end{aligned}$$

(b):

$$\begin{aligned}\nabla (\vec{v} \cdot \vec{w}) &= \frac{\partial}{\partial x_i} (\vec{v} \cdot \vec{w}) \vec{e}_i \\ &= \frac{\partial}{\partial x_i} (v_k w_k) \vec{e}_i \\ &= \left(\frac{\partial v_k}{\partial x_i} w_k + v_k \frac{\partial w_k}{\partial x_i} \right) \vec{e}_i\end{aligned}$$

$$\begin{aligned}
&= \left([\nabla \vec{v}^T]_{ik} w_k + [\nabla \vec{w}^T]_{ik} v_k \right) \vec{e}_i \\
&= \left([\nabla v^T \vec{w}]_i + [\nabla w^T \vec{v}]_i \right) \vec{e}_i \\
&= \nabla \vec{v}^T \vec{w} + \nabla \vec{w}^T \vec{v} //
\end{aligned}$$

(c)

$$\begin{aligned}
\nabla \cdot (\vec{v} \otimes \vec{w}) &= \frac{\partial}{\partial x_j} (v_i w_j) \vec{e}_i \\
&= \left(\frac{\partial v_i}{\partial x_j} w_j + v_i \frac{\partial w_j}{\partial x_j} \right) \vec{e}_i \\
&= [\nabla \vec{v}] \vec{w} + \vec{v} \nabla \cdot \vec{w} \\
&= (\nabla \vec{v}) \vec{w} + (\nabla \cdot \vec{w}) \vec{v} //
\end{aligned}$$